Bagged Random Subset Forecast Combination under Common Loading

A Small-Heterogeneity Derivation of the Bagged Approximation Error, Optimal Subset Size, and Dominance Regions

RSM Forecast-Combination Project

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Abstract

This note gives a focused theory for Random Subset Methods (RSM) in the common-loading heteroskedastic forecast-error model. It removes the earlier practice of postulating a bagged approximation error of the form C/k^2. Instead, it derives an endpoint-corrected squared approximation from a small-heterogeneity expansion of idiosyncratic precisions. If $\tau_i = \bar{\tau}(1+\delta_i)$ with $m^{-1}\sum_i \delta_i = 0$ and small dispersion $V_\delta = m^{-1}\sum_i \delta_i^2$, then the infinite-bagged excess risk satisfies

$$Q_k^{\text{bag},\infty} - Q^{\text{opt}} \approx C_\delta \left(\frac{1}{k} - \frac{1}{m}\right)^2, \qquad C_\delta = \frac{BV_\delta}{m}\left(\frac{m}{m-1}\right)^2, \quad B = \frac{1}{\bar{\tau}}.$$

This approximation is derived from the bagged weights, is valid as a local small-heterogeneity statement, and satisfies the exact boundary condition $Q_k^{\text{bag},\infty}\big|_{k=m} = Q^{\text{opt}}$. For $k \ll m$ it reduces to the local form C_δ/k^2, which yields the cube-root rule

$$k^*_{\text{bag}} \asymp \left(\frac{T}{m}\right)^{1/3}$$

when precision heterogeneity and the irreducible/common-loading loss component are stable. The note also derives finite-bagging corrections, showing how a finite number of bags reintroduces a first-order term proportional to $1/k - 1/m$, and establishes dominance regions against full estimated optimal weights, equal weights, and generic benchmarks.

Contents

1 Scope and main message

The purpose of this note is to focus only on the common-loading forecast-error model. We keep the distinction between two RSM objects:

$$Q_k^{\text{sub}} = E_S[v_S' \Sigma_e v_S], \tag{1}$$

$$Q_k^{\text{bag},\infty} = (E_S v_S)' \Sigma_e (E_S v_S), \tag{2}$$

where v_S is the full m-dimensional vector that embeds the optimal weights computed within subset S and puts zeros outside S.

The first object is the average loss of a randomly selected subset forecast. The second object is the loss of the infinitely bagged forecast, where subset forecasts are averaged first and the loss is evaluated afterward. Since $w \mapsto w' \Sigma_e w$ is convex,

$$Q_k^{\text{bag},\infty} \le Q_k^{\text{sub}}.$$

Thus average-subset theory is a conservative upper-bound theory for fully bagged RSM, but it is generally not the exact population theory of the practical bagged estimator.

The key contribution of this note is the following. Instead of assuming

$$Q_k^{\text{bag},\infty} - Q^{\text{opt}} \approx \frac{C}{k^2},$$

we derive the endpoint-corrected approximation

$$Q_k^{\text{bag},\infty} - Q^{\text{opt}} \approx C_\delta \left(\frac{1}{k} - \frac{1}{m} \right)^2 \tag{3}$$

from a small-heterogeneity expansion of the precision sequence $\{\tau_i\}_{i=1}^m$. This route avoids relying on a large-k expansion when the empirical optimal k is small.

2 Forecast-combination setup

There are m individual forecasts

$$f_t = (f_{1t}, \dots, f_{mt})'.$$

of a scalar target

$$y_t = \mu_t + \varepsilon_t.$$

Define forecast errors relative to the predictable target component by

$$u_t = f_t - \mu_t \iota_m.$$

Assume

$$E(\varepsilon_t) = 0, \quad \text{Var}(\varepsilon_t) = s > 0, \quad E(\varepsilon_t u_{it}) = 0,$$

and

$$E(u_t) = 0, \quad \text{Var}(u_t) = \Sigma_e \succ 0.$$

For any weight vector w satisfying $w'\iota_m = 1$,

$$w'f_t - y_t = w'u_t - \varepsilon_t.$$

Therefore,

$$E[(w'f_t - y_t)^2] = s + w'\Sigma_e w.$$

The term s is common to all forecast combinations. Population differences across combinations are governed by

$$Q(w) = w'\Sigma_e w.$$

3 Benchmarks

3.1 Equal weights

Equal weights use

$$w_{\text{ave}} = \frac{1}{m}\iota_m.$$

Their population forecast-error component is

$$Q^{\text{ave}} = \frac{1}{m^2}\iota_m'\Sigma_e\iota_m.$$

Since no combination weights are estimated, the benchmark loss is

$$L^{\text{ave}} = s + Q^{\text{ave}}.$$

3.2 Full optimal weights

The population optimal combination solves

$$\min_w \; w'\Sigma_e w \quad \text{s.t.} \quad w'\iota_m = 1.$$

The solution is

$$w^{\text{opt}} = \frac{\Sigma_e^{-1}\iota_m}{\iota_m'\Sigma_e^{-1}\iota_m}, \qquad Q^{\text{opt}} = (\iota_m'\Sigma_e^{-1}\iota_m)^{-1}.$$

If the full optimal weights are estimated from T observations, the usual degrees-of-freedom approximation gives

$$L^{\text{opt}} \approx (s + Q^{\text{opt}})\frac{T-1}{T-m}, \qquad T > m.$$

4 Subset weights and the two RSM objects

Let $S \subset \{1, \dots, m\}$ be a subset of size k. Let $\Sigma_{e,S}$ denote the $k \times k$ principal submatrix of Σ_e indexed by S. Inside subset S, the population optimal subset weights are

$$w_S = \frac{\Sigma_{e,S}^{-1}\iota_S}{\iota_S'\Sigma_{e,S}^{-1}\iota_S}.$$

Embed w_S into $\mathbb{R}^m$ by defining $v_S = P_S w_S$, where P_S places the k selected weights in their original coordinates and sets all other coordinates to zero. Then $v_S'\iota_m = 1$.

4.1 Average-subset RSM

The average-subset population risk is

$$Q_k^{\text{sub}} = E_S[v_S' \Sigma_e v_S] = E_S[Q(S)],$$

where

$$Q(S) = (\iota_S' \Sigma_{e,S}^{-1} \iota_S)^{-1}.$$

This object corresponds to computing the loss of a single random subset forecast and then averaging those losses across random subsets.

4.2 Infinite-bagged RSM

The infinitely bagged RSM weight vector is

$$\bar{v}_k = E_S[v_S].$$

The infinite-bagged population risk is

$$Q_k^{\text{bag},\infty} = \bar{v}_k' \Sigma_e \bar{v}_k.$$

This is the population object approached when many random subset forecasts are averaged before computing the forecast loss.

4.3 Finite number of bagging draws

Let $S_1, \ldots, S_G$ be independent random subsets of size k and define

$$\hat{v}_{k,G} = \frac{1}{G} \sum_{g=1}^{G} v_{S_g}.$$

The finite-bagged population risk, averaging over subset draws, is

$$Q_k^{\text{bag},G} = E_S[\hat{v}_{k,G}' \Sigma_e \hat{v}_{k,G}].$$

Proposition 1 (Finite-bagging bridge). *For independent subset draws,*

$$Q_k^{\text{bag},G} = Q_k^{\text{bag},\infty} + \frac{1}{G}\big(Q_k^{\text{sub}} - Q_k^{\text{bag},\infty}\big). \tag{4}$$

Consequently,

$$Q_k^{\text{bag},\infty} \le Q_k^{\text{bag},G} \le Q_k^{\text{sub}}.$$

Proof. Let $V = v_S$ and $\bar{v} = E[V]$. For independent draws, $\hat{v} = G^{-1} \sum_{g=1}^{G} V_g$. Then

$$\begin{aligned} E[\hat{v}' \Sigma_e \hat{v}] &= \bar{v}' \Sigma_e \bar{v} + E[(\hat{v} - \bar{v})' \Sigma_e (\hat{v} - \bar{v})] \\ &= \bar{v}' \Sigma_e \bar{v} + \frac{1}{G} E[(V - \bar{v})' \Sigma_e (V - \bar{v})] \\ &= Q_k^{\text{bag},\infty} + \frac{1}{G}(Q_k^{\text{sub}} - Q_k^{\text{bag},\infty}). \end{aligned}$$

The inequalities follow from $\Sigma_e \succ 0$. $\square$

5 Finite-sample inflation

For a single subset with k forecasts, constrained least squares has $k-1$ free combination parameters. Under the Gaussian random-design approximation, the out-of-sample inflation factor is

$$I(T,k) = \frac{T-1}{T-k}.$$

Thus the average-subset loss approximation is

$$L_k^{\mathrm{sub}} \approx (s + Q_k^{\mathrm{sub}})\frac{T-1}{T-k}.$$

For bagged RSM, the exact effective degrees of freedom can be smaller because averaging overlapping subset estimators reduces variance. The conservative comparable approximation is

$$L_k^{\mathrm{bag}} \approx (s + Q_k^{\mathrm{bag}})\frac{T-1}{T-k}.$$

A more general version replaces k with an effective degree count $d^{\mathrm{bag}}(k)$, but the conservative calculations below use $d^{\mathrm{bag}}(k) = k$.

6 Common-loading heteroskedastic forecast-error model

6.1 Model

Assume forecast errors satisfy

$$u_t = \lambda F_t + \xi_t,$$

where all forecasts have the same factor loading λ, $\mathrm{Var}(\lambda F_t) = q > 0$, and

$$\mathrm{Var}(\xi_t) = D = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_m^2).$$

Define idiosyncratic precisions

$$\tau_i = \sigma_i^{-2} > 0, \qquad D_{ii} = \sigma_i^2 = \frac{1}{\tau_i}.$$

Then

$$\Sigma_e = q\iota_m\iota_m' + D.$$

For any feasible weight vector $w'\iota_m = 1$,

$$Q(w) = w'\Sigma_e w = q + \sum_{i=1}^{m} \frac{w_i^2}{\tau_i}.$$

The common component q cannot be diversified away because all weights sum to one and all forecasts have the same common loading.

6.2 Exact subset risk

For subset S,

$$\Sigma_{e,S} = q\iota_S\iota_S' + D_S.$$

Let

$$A_S = \sum_{i\in S}\tau_i.$$

Using the Sherman–Morrison formula,

$$\Sigma_{e,S}^{-1} = D_S^{-1} - \frac{qD_S^{-1}\iota_S\iota_S' D_S^{-1}}{1+q\iota_S' D_S^{-1}\iota_S}.$$

Since $\iota_S' D_S^{-1}\iota_S = A_S$,

$$\iota_S'\Sigma_{e,S}^{-1}\iota_S = A_S - \frac{qA_S^2}{1+qA_S} = \frac{A_S}{1+qA_S}.$$

Therefore,

$$Q(S) = (\iota_S'\Sigma_{e,S}^{-1}\iota_S)^{-1} = q + \frac{1}{A_S}.$$

The subset optimal weights are

$$w_{i,S} = \frac{\tau_i}{A_S}, \quad i \in S.$$

Thus

$$Q_k^{\text{sub}} = q + E_S\left[\frac{1}{\sum_{i\in S}\tau_i}\right].$$

6.3 Full optimal weights

For the full pool, $A_m = \sum_{i=1}^m \tau_i$. The full optimal weights are

$$w_i^{\text{opt}} = \frac{\tau_i}{A_m},$$

and the optimal population risk is

$$Q^{\text{opt}} = q + \frac{1}{A_m}.$$

6.4 Exact infinite-bagged expression

For a fixed subset S,

$$v_{S,i} = \mathbf{1}\{i\in S\}\frac{\tau_i}{A_S}.$$

Therefore the infinite-bagged weight on forecast i is

$$\bar{v}_{i,k} = E_S[v_{S,i}] = \tau_i\, E_R\left[\frac{k/m}{\tau_i + \sum_{j\in R}\tau_j}\right], \tag{5}$$

where R is drawn uniformly from all subsets of $\{1,\dots,m\}\setminus\{i\}$ of size $k-1$. Since $\sum_i \bar v_{i,k} = 1$, the common factor contributes q. Hence

$$Q_k^{\mathrm{bag},\infty} = q + \sum_{i=1}^{m} \frac{\bar v_{i,k}^2}{\tau_i}.$$

This exact expression is valid for every k, including small k.

6.5 Decomposition around full optimal weights

Let

$$\Delta_{i,k} = \bar v_{i,k} - w_i^{\mathrm{opt}}.$$

Because both $\bar v_k$ and w^{opt} sum to one, $\sum_i \Delta_{i,k} = 0$. Also, $w_i^{\mathrm{opt}}/\tau_i = 1/A_m$. Therefore,

$$\begin{aligned} Q_k^{\mathrm{bag},\infty} &= q + \sum_i \frac{(w_i^{\mathrm{opt}} + \Delta_{i,k})^2}{\tau_i} \\ &= q + \sum_i \frac{(w_i^{\mathrm{opt}})^2}{\tau_i} + \frac{2}{A_m}\sum_i \Delta_{i,k} + \sum_i \frac{\Delta_{i,k}^2}{\tau_i} \\ &= Q^{\mathrm{opt}} + \sum_i \frac{\Delta_{i,k}^2}{\tau_i}. \end{aligned}$$

Thus

$$R_k^{\mathrm{bag},\infty} := Q_k^{\mathrm{bag},\infty} - Q^{\mathrm{opt}} = \sum_{i=1}^{m} \frac{(\bar v_{i,k} - w_i^{\mathrm{opt}})^2}{\tau_i} \geq 0. \tag{6}$$

Remark 1. *The excess bagged risk is not first order in the weight error. The linear term vanishes because the full optimal weights satisfy the first-order condition under the adding-up constraint. Hence the excess risk is a weighted squared distance between $\bar v_k$ and w^{opt}.*

7 Small-heterogeneity derivation of the bagged approximation error

7.1 Small-heterogeneity assumption

Instead of assuming a functional form for $R_k^{\mathrm{bag},\infty}$, assume that idiosyncratic precisions are close to homogeneous:

$$\tau_i = \bar\tau(1+\delta_i), \qquad \frac{1}{m}\sum_i \delta_i = 0, \qquad \max_i |\delta_i| \text{ is small.}$$

Define

$$B = \frac{1}{\bar\tau}, \qquad V_\delta = \frac{1}{m}\sum_i \delta_i^2.$$

Then

$$A_m = m\bar\tau, \qquad w_i^{\mathrm{opt}} = \frac{1+\delta_i}{m}, \qquad Q^{\mathrm{opt}} = q + \frac{B}{m}.$$

7.2 Expansion of average-subset risk

For a subset S, write

$$A_S = \sum_{i \in S} \tau_i = \bar{\tau}(k + \Delta_S), \qquad \Delta_S = \sum_{i \in S} \delta_i.$$

Under simple random sampling without replacement,

$$E[\Delta_S] = 0, \qquad \mathrm{Var}(\Delta_S) = \frac{k(m-k)}{m-1} V_\delta.$$

Using the second-order Taylor expansion of $1/(k + \Delta_S)$, we obtain

$$Q_k^{\mathrm{sub}} \approx q + \frac{B}{k} + B \frac{m-k}{k^2(m-1)} V_\delta. \tag{7}$$

Thus the average-subset object has a leading first-order term B/k. The heterogeneity correction is of smaller order in δ.

7.3 Expansion of bagged weights

The exact infinite-bagged weight is

$$\bar{v}_{i,k} = \tau_i \, E_R \left[\frac{k/m}{\tau_i + \sum_{j \in R} \tau_j} \right].$$

Using $\tau_i = \bar{\tau}(1 + \delta_i)$ and writing $\Delta_R = \sum_{j \in R} \delta_j$, we get

$$\bar{v}_{i,k} = \frac{1 + \delta_i}{m} \, E_R \left[\frac{k}{k + \delta_i + \Delta_R} \right].$$

Conditional on i being included, R is drawn from the remaining $m-1$ units. Since $\sum_{j \neq i} \delta_j = -\delta_i$,

$$E_R[\Delta_R] = \frac{k-1}{m-1}(-\delta_i).$$

Therefore,

$$\delta_i + E_R[\Delta_R] = \delta_i \frac{m-k}{m-1}.$$

Expanding the reciprocal to first order in δ gives

$$E_R \left[\frac{1}{k + \delta_i + \Delta_R} \right] = \frac{1}{k} - \frac{\delta_i(m-k)}{k^2(m-1)} + O(\delta^2).$$

Thus

$$\bar{v}_{i,k} - w_i^{\mathrm{opt}} = -\frac{\delta_i}{m} \cdot \frac{m-k}{k(m-1)} + O(\delta^2). \tag{8}$$

This expression is first order in heterogeneity but does not require k to be large.

7.4 Endpoint-corrected squared approximation

The bagged excess risk is

$$R_k^{\text{bag},\infty} = \sum_i \frac{(\bar{v}_{i,k} - w_i^{\text{opt}})^2}{\tau_i}.$$

Because $1/\tau_i = B(1 + O(\delta_i))$, and $(\bar{v}_{i,k} - w_i^{\text{opt}})^2$ is already second order in δ_i,

$$R_k^{\text{bag},\infty} \approx B \sum_i \left(\frac{\delta_i}{m} \cdot \frac{m-k}{k(m-1)} \right)^2 = \frac{BV_\delta}{m} \left(\frac{m-k}{k(m-1)} \right)^2.$$

Since

$$\frac{m-k}{k(m-1)} = \frac{m}{m-1} \left(\frac{1}{k} - \frac{1}{m} \right),$$

we have the endpoint-corrected approximation

$$R_k^{\text{bag},\infty} \approx C_\delta \left(\frac{1}{k} - \frac{1}{m} \right)^2, \qquad C_\delta = \frac{BV_\delta}{m} \left(\frac{m}{m-1} \right)^2. \tag{9}$$

Therefore,

$$Q_k^{\text{bag},\infty} \approx q + \frac{B}{m} + C_\delta \left(\frac{1}{k} - \frac{1}{m} \right)^2.$$

Remark 2. *The squared dependence on $1/k - 1/m$ is not imposed as a primitive assumption. It follows from two facts: first, under small precision heterogeneity, the bagged weight error is proportional to $(1/k - 1/m)\delta_i$; second, excess risk is quadratic in the weight error around the full optimum. For $k \ll m$, the endpoint-corrected term behaves like C_δ/k^2, but the exact endpoint $R_m^{\text{bag},\infty} = 0$ is preserved.*

Remark 3 (Homoskedastic special case)**.** *If $\delta_i = 0$ for all i, then $V_\delta = 0$, $C_\delta = 0$, and*

$$Q_k^{\text{bag},\infty} = Q^{\text{opt}}$$

for every k. Bagging fully removes the population dependence on k in the homoskedastic common-loading model.

8 Finite bagging under small heterogeneity

By the finite-bagging bridge,

$$Q_k^{\text{bag},G} = Q_k^{\text{bag},\infty} + \frac{1}{G}(Q_k^{\text{sub}} - Q_k^{\text{bag},\infty}).$$

Using the leading average-subset approximation $Q_k^{\text{sub}} - Q^{\text{opt}} \approx B(1/k - 1/m)$ and the small-heterogeneity bagged approximation above, we obtain

$$R_k^{\text{bag},G} \approx \frac{B}{G} \left(\frac{1}{k} - \frac{1}{m} \right) + \left(1 - \frac{1}{G} \right) C_\delta \left(\frac{1}{k} - \frac{1}{m} \right)^2. \tag{10}$$

This expression shows that finite bagging reintroduces a first-order term. Infinite bagging removes the $1/k$ component; finite bagging keeps a fraction $1/G$ of it.

For compact notation, define $z_k = 1/k - 1/m$, $A = s + Q^{\mathrm{opt}} = s + q + B/m$, $a_G = B/G$, $c_G = (1 - 1/G)C_\delta$. Then

$$Q_k^{\mathrm{bag},G} \approx Q^{\mathrm{opt}} + a_G z_k + c_G z_k^2. \tag{11}$$

9 Optimal subset size

9.1 Average-subset optimum

Using the leading average-subset approximation $Q_k^{\mathrm{sub}} \approx q + B/k$, define $A_0 = s + q$. The finite-sample loss is

$$L_k^{\mathrm{sub}} \approx \left(A_0 + \frac{B}{k}\right)\frac{T-1}{T-k}.$$

The first-order condition gives

$$A_0 k^2 + 2Bk - BT = 0,$$

with positive root

$$k_{\mathrm{sub}}^* = \frac{-B + \sqrt{B^2 + A_0 BT}}{A_0} \sim \left(\frac{BT}{A_0}\right)^{1/2}. \tag{12}$$

Thus the average-subset object has the familiar square-root rule.

9.2 Infinite-bagged optimum: endpoint-corrected expression

Using the small-heterogeneity bagged approximation,

$$L_k^{\mathrm{bag},\infty} \approx \left(A + C_\delta z_k^2\right)\frac{T-1}{T-k}, \qquad z_k = \frac{1}{k} - \frac{1}{m}.$$

The first-order condition for an interior optimum is

$$A + C_\delta z_k^2 - \frac{2C_\delta z_k}{k^2}(T-k) = 0.$$

Equivalently,

$$Am^2k^3 + C_\delta k(m-k)^2 - 2C_\delta m(m-k)(T-k) = 0. \tag{13}$$

This condition can be solved numerically over the integer grid $k = 1, \ldots, m-1$. If $k \ll m$, then $z_k \approx 1/k$, and the first-order condition reduces to $Ak^3 + 3C_\delta k - 2C_\delta T = 0$. For large T and interior k,

$$k_{\mathrm{bag},\infty}^* \sim \left(\frac{2C_\delta T}{A}\right)^{1/3}. \tag{14}$$

Substituting C_δ gives

$$k_{\mathrm{bag},\infty}^* \sim \left[\frac{2BV_\delta T}{m(s+q+B/m)}\left(\frac{m}{m-1}\right)^2\right]^{1/3}.$$

Therefore, if V_δ, B, and $s+q$ are stable as T and m vary,

$$k^*_{\mathrm{bag},\infty} \propto \left(\frac{T}{m}\right)^{1/3}. \tag{15}$$

Remark 4. *The endpoint-corrected approximation above is derived from small heterogeneity, not from a large-k expansion. Therefore it can be used to interpret small optimal k more safely than the raw local approximation C/k^2. The raw cube-root rule is still a local simplification when $k \ll m$.*

9.3 Finite-bagged optimum

Using $Q_k^{\mathrm{bag},G} \approx Q^{\mathrm{opt}} + a_G z_k + c_G z_k^2$, the finite-bagged loss is

$$L_k^{\mathrm{bag},G} \approx (A + a_G z_k + c_G z_k^2)\frac{T-1}{T-k}.$$

The first-order condition for $k \ll m$ reduces to

$$Ak^3 + 2a_G k^2 + (3c_G - a_G T)k - 2c_G T = 0. \tag{16}$$

This nests two limiting regimes:

1. If G is large, then $a_G = B/G$ is small and the squared bagged term dominates. Then

$$k^*_{\mathrm{bag},G} \approx \left(\frac{2c_G T}{A}\right)^{1/3}.$$

2. If G is small enough that the first-order finite-bagging term dominates, then

$$k^*_{\mathrm{bag},G} \approx \left(\frac{a_G T}{A}\right)^{1/2} = \left(\frac{BT}{GA}\right)^{1/2}.$$

Thus finite bagging creates a transition between a cube-root rule and a square-root rule.

10 Dominance regions

This section establishes dominance regions using the common-loading small-heterogeneity approximation. We treat k as an integer in $\{1,\ldots,m-1\}$ and require positive denominators such as $T > k$ and, when comparing against full estimated optimal weights, $T > m$.

10.1 Average subset versus full estimated optimal weights

Let $A_0 = s + q$. The leading average-subset loss is

$$L_k^{\mathrm{sub}} \approx \left(A_0 + \frac{B}{k}\right)\frac{T-1}{T-k}.$$

The full estimated optimal-weight benchmark is

$$L^{\text{opt}} \approx \left(A_0 + \frac{B}{m}\right) \frac{T-1}{T-m}.$$

Assuming $T > m$, average-subset RSM beats full estimated optimal weights iff

$$\left(A_0 + \frac{B}{k}\right) \frac{1}{T-k} < \left(A_0 + \frac{B}{m}\right) \frac{1}{T-m}.$$

This is equivalent to

$$\frac{B(T-m)}{mA_0 + B} < k < m. \tag{17}$$

10.2 Average subset versus equal weights

In the common-loading model,

$$Q^{\text{ave}} = q + \frac{1}{m^2} \sum_{i=1}^{m} \sigma_i^2.$$

Let $M_{\text{ave}} = s + Q^{\text{ave}}$. Average-subset RSM beats equal weights iff

$$\left(A_0 + \frac{B}{k}\right) \frac{T-1}{T-k} < M_{\text{ave}}.$$

Canceling $T-1$ and rearranging yields the quadratic inequality

$$M_{\text{ave}} k^2 - P_{\text{ave}} k + B(T-1) < 0, \qquad P_{\text{ave}} = M_{\text{ave}} T - A_0(T-1).$$

If $\Delta_{\text{ave}} = P_{\text{ave}}^2 - 4M_{\text{ave}} B(T-1) > 0$, then the dominance interval is $\kappa_{\text{ave}}^- < k < \kappa_{\text{ave}}^+$, where

$$\kappa_{\text{ave}}^{\pm} = \frac{P_{\text{ave}} \pm \sqrt{\Delta_{\text{ave}}}}{2M_{\text{ave}}}.$$

10.3 Infinite-bagged RSM versus full estimated optimal weights

Let $A = s + Q^{\text{opt}} = s + q + B/m$. The infinite-bagged loss approximation is

$$L_k^{\text{bag},\infty} \approx (A + C_\delta z_k^2) \frac{T-1}{T-k}, \qquad z_k = \frac{1}{k} - \frac{1}{m}.$$

The full estimated optimal loss is $L^{\text{opt}} \approx A(T-1)/(T-m)$. For $T > m$, bagged RSM beats full estimated optimal weights iff

$$(A + C_\delta z_k^2) \frac{1}{T-k} < \frac{A}{T-m}.$$

This is equivalent to

$$C_\delta z_k^2 (T-m) < A(m-k).$$

Using $z_k^2 = (m-k)^2/(k^2m^2)$ and dividing by $m-k>0$,

$$Am^2k^2 > C_\delta(T-m)(m-k). \tag{18}$$

Equivalently, $Am^2k^2 + C_\delta(T-m)k - C_\delta m(T-m) > 0$. The dominance region is $\kappa^{\mathrm{opt},\infty} < k < m$, where

$$\kappa^{\mathrm{opt},\infty} = \frac{-C_\delta(T-m) + \sqrt{C_\delta^2(T-m)^2 + 4AC_\delta m^3(T-m)}}{2Am^2}. \tag{19}$$

10.4 Finite-bagged RSM versus full estimated optimal weights

For finite G,

$$L_k^{\mathrm{bag},G} \approx (A + a_G z_k + c_G z_k^2)\frac{T-1}{T-k}.$$

Comparing to $L^{\mathrm{opt}} = A(T-1)/(T-m)$, finite-bagged RSM beats full estimated optimal weights iff

$$Am^2k^2 > (T-m)\big[a_G mk + c_G(m-k)\big]. \tag{20}$$

This condition reduces to (18) when $a_G = 0$ and $c_G = C_\delta$.

10.5 Infinite-bagged RSM versus a generic benchmark

Let a benchmark have population component Q_0 and effective degree count d_0, so that $L_0 = (s+Q_0)(T-1)/(T-d_0)$. Define

$$M_0 = s + Q_0, \quad A = s + Q^{\mathrm{opt}}, \quad P_0 = M_0T - A(T-d_0).$$

Infinite-bagged RSM beats this benchmark iff $F_\infty(k) < 0$, where

$$F_\infty(k) = M_0m^2k^3 - P_0m^2k^2 + C_\delta(T-d_0)(m-k)^2. \tag{21}$$

For equal weights, set $Q_0 = Q^{\mathrm{ave}}$ and $d_0 = 0$. For full estimated optimal weights, set $Q_0 = Q^{\mathrm{opt}}$ and $d_0 = m$.

10.6 Finite-bagged RSM versus a generic benchmark

For finite G,

$$F_G(k) = M_0m^2k^3 - P_0m^2k^2 + (T-d_0)\big[a_G mk(m-k) + c_G(m-k)^2\big].$$

The dominance region is the set of integers $k \in \{1,\dots,m-1\}$ satisfying $F_G(k) < 0$.

11 How to use this in simulations

The Monte Carlo design should report separate objects:

1. The exact common-loading bagged population curve:

$$Q_k^{\text{bag},\infty} = q + \sum_{i=1}^{m} \frac{\bar{v}_{i,k}^2}{\tau_i}, \qquad \bar{v}_{i,k} = \tau_i \, E_R\left[\frac{k/m}{\tau_i + \sum_{j\in R}\tau_j}\right].$$

   This is the most reliable population benchmark, especially when the optimal k is small.

2. The small-heterogeneity approximation:

$$Q_k^{\text{bag},\infty} \approx q + \frac{B}{m} + C_\delta \left(\frac{1}{k} - \frac{1}{m}\right)^2.$$

   This is the main interpretable approximation because it is derived from the precision heterogeneity and respects the endpoint at $k = m$.

3. The finite-bagging approximation:

$$Q_k^{\text{bag},G} \approx q + \frac{B}{m} + \frac{B}{G}\left(\frac{1}{k} - \frac{1}{m}\right) + \left(1 - \frac{1}{G}\right) C_\delta \left(\frac{1}{k} - \frac{1}{m}\right)^2.$$

   This shows whether the finite number of bags is large enough for the infinite-bagged cube-root logic to dominate.

4. The log-log scaling check: $\log k^* = \alpha + \beta_T \log T + \beta_m \log m + u$. The infinite-bagged small-heterogeneity theory predicts approximately $\beta_T \approx 1/3$, $\beta_m \approx -1/3$, provided B, V_δ, and $s + q$ are stable across grid points.

12 Conclusion

In the common-loading heteroskedastic forecast-error model, average-subset and bagged RSM are different objects. The average-subset risk satisfies

$$Q_k^{\text{sub}} = q + E_S\left[\frac{1}{\sum_{i\in S}\tau_i}\right] \approx q + \frac{B}{k},$$

which leads to a square-root optimal subset-size rule.

The infinitely bagged object is

$$Q_k^{\text{bag},\infty} = q + \sum_{i=1}^{m} \frac{\bar{v}_{i,k}^2}{\tau_i} = Q^{\text{opt}} + R_k^{\text{bag},\infty},$$

where

$$R_k^{\text{bag},\infty} = \sum_{i=1}^{m} \frac{(\bar{v}_{i,k} - w_i^{\text{opt}})^2}{\tau_i}.$$

Under small precision heterogeneity, $\tau_i = \bar{\tau}(1+\delta_i)$, $V_\delta = m^{-1}\sum_i \delta_i^2$, the bagged excess risk satisfies

$$R_k^{\text{bag},\infty} \approx \frac{BV_\delta}{m}\left(\frac{m}{m-1}\right)^2\left(\frac{1}{k}-\frac{1}{m}\right)^2.$$

This derives the squared form from the behavior of the bagged weights rather than assuming it. When $k \ll m$, the approximation reduces to a local C/k^2 form and implies

$$k^*_{\text{bag},\infty} \sim \left[\frac{2BV_\delta T}{m(s+q+B/m)}\left(\frac{m}{m-1}\right)^2\right]^{1/3}.$$

Thus, under stable heterogeneity and stable irreducible/common-loading loss, $k^*_{\text{bag},\infty} \propto (T/m)^{1/3}$.

For finite bagging, a first-order term remains:

$$R_k^{\text{bag},G} \approx \frac{B}{G}\left(\frac{1}{k}-\frac{1}{m}\right) + \left(1-\frac{1}{G}\right)C_\delta\left(\frac{1}{k}-\frac{1}{m}\right)^2,$$

so the optimal k may transition toward a square-root rule when G is small.

The dominance results show explicitly when bagged RSM can beat full estimated optimal weights or equal weights. In particular, infinite-bagged RSM beats full estimated optimal weights when

$$Am^2k^2 > C_\delta(T-m)(m-k), \qquad A = s+q+B/m,$$

while finite-bagged RSM beats full estimated optimal weights when

$$Am^2k^2 > (T-m)\big[a_G mk + c_G(m-k)\big].$$

These formulas make clear why bagged RSM may be attractive at low subset sizes: the population bagged curve can become close to the full optimum quickly, while the finite-sample inflation still penalizes large k.

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Appendix: Exact Dominance Regions under Common Loading

Supplement to "Bagged Random Subset Forecast Combination under Common Loading"

May 2026

Overview. The dominance conditions in Section 10 of the main text are derived under the small-heterogeneity approximation. This appendix shows that the common-loading model already yields *exact* dominance conditions—no additional assumption on the dispersion of $\{\tau_i\}$ is needed. The small-heterogeneity expressions in Section 10 are then recovered as corollaries by substituting the approximate forms of Q_k^{sub} and $R_k^{\text{bag},\infty}$.

11 Exact Dominance Regions under Common Loading

11.1 Setup and key exact identities

Recall the common-loading model $\Sigma_e = q\iota_m\iota_m' + D$, $D = \text{diag}(\sigma_1^2, \ldots, \sigma_m^2)$, $\tau_i = \sigma_i^{-2}$. Define

$$A_S = \sum_{i \in S} \tau_i, \qquad A_m = \sum_{i=1}^{m} \tau_i, \qquad B = \tfrac{1}{\bar{\tau}} = \tfrac{m}{A_m}.$$

The following three identities are exact in the common-loading model, without any heterogeneity assumption.

1. **Subset risk:**
$$Q(S) = q + \tfrac{1}{A_S}, \qquad Q_k^{\text{sub}} = q + E_S\left[\tfrac{1}{A_S}\right]. \tag{164}$$

2. **Optimal risk and weights:**
$$Q^{\text{opt}} = q + \tfrac{1}{A_m}, \qquad w_i^{\text{opt}} = \tfrac{\tau_i}{A_m}. \tag{165}$$

3. **Exact excess bagged risk (from eq. 60 of the main text):**
$$R_k^{\text{bag},\infty} := Q_k^{\text{bag},\infty} - Q^{\text{opt}} = \sum_{i=1}^{m} \frac{(\bar{v}_{i,k} - w_i^{\text{opt}})^2}{\tau_i} \;\geq\; 0. \tag{166}$$

   This identity is exact for every $k \in \{1, \ldots, m\}$ and every precision sequence $\{\tau_i\}$ with $\tau_i > 0$. The boundary conditions $R_m^{\text{bag},\infty} = 0$ (bagging the full pool recovers the optimum) and $R_k^{\text{bag},\infty} \geq 0$ both hold by construction.

For the finite-sample comparisons we use the inflation factor $I(T,k) = (T-1)/(T-k)$ from Section 5, with the conservative convention $d^{\text{bag}}(k) = k$. Define:

$$A := s + Q^{\text{opt}} = s + q + \tfrac{1}{A_m}, \qquad M_{\text{ave}} := s + Q^{\text{ave}} = s + q + \tfrac{1}{m^2}\sum_{i=1}^{m} \sigma_i^2. \tag{167}$$

Note $A \le M_{\text{ave}}$ by the optimality of w^{opt}.

11.2 Average-subset RSM versus full estimated optimal weights

Proposition 2 (Exact average-subset dominance)**.** *Assume $T > m$ so that the full estimated weights are feasible. Average-subset RSM beats full estimated optimal weights at subset size k iff*

$$\boxed{\left(Q_k^{\text{sub}} - Q^{\text{opt}}\right)(T-m) \;<\; A\,(m-k).} \tag{168}$$

Equivalently, using $Q_k^{\text{sub}} = q + E_S[1/A_S]$ and $Q^{\text{opt}} = q + 1/A_m$,

$$\left(E_S\left[\frac{1}{A_S}\right] - \frac{1}{A_m}\right)(T-m) \;<\; \left(s + q + \frac{1}{A_m}\right)(m-k). \tag{169}$$

Proof. The average-subset loss is $L_k^{\text{sub}} = (s + Q_k^{\text{sub}})\frac{T-1}{T-k}$ and the full optimal benchmark is $L^{\text{opt}} = (s + Q^{\text{opt}})\frac{T-1}{T-m}$. Cancel $T - 1 > 0$ and cross-multiply by $(T-k)(T-m) > 0$:

$$(s + Q_k^{\text{sub}})(T-m) < (s + Q^{\text{opt}})(T-k).$$

Write $s + Q_k^{\text{sub}} = A + (Q_k^{\text{sub}} - Q^{\text{opt}})$ and expand:

$$A(T-m) + (Q_k^{\text{sub}} - Q^{\text{opt}})(T-m) < A(T-k).$$

Rearranging gives (168). □

Remark 1. *The dominance condition (168) depends on the precision sequence only through $E_S[1/A_S]$ and A_m. The quantity $E_S[1/A_S]$ is monotone decreasing in k (larger subsets have smaller average subset risk) and equals $1/A_m$ at $k = m$. Hence there exists a threshold $k^\dagger$ such that (168) holds for all $k \in (k^\dagger, m)$. The threshold can be computed numerically via Monte Carlo estimation of $E_S[1/A_S]$ (see Section 11 of the main text), without any parametric assumption on $\{\tau_i\}$.*

Corollary 1 (Small-heterogeneity recovery)**.** *Under the small-heterogeneity approximation $E_S[1/A_S] \approx B/k$, condition (168) reduces to*

$$\frac{B(T-m)}{km} < A,$$

i.e. $k > B(T-m)/(mA)$, which matches eq. (117) of the main text.

11.3 Average-subset RSM versus equal weights

Equal weights require no estimation, so their loss is $L^{\text{ave}} = s + Q^{\text{ave}}$ (no inflation).

Proposition 3 (Exact average-subset vs equal-weights dominance)**.** *Average-subset RSM beats equal weights at subset size k iff*

$$\boxed{\left(s + q + E_S\left[\frac{1}{A_S}\right]\right)(T-1) \;<\; M_{\text{ave}}\,(T-k).} \tag{170}$$

This holds whenever the right-hand side exceeds the left-hand side, which requires

$$T \;>\; \frac{M_{\text{ave}}\,k \;-\; (s + Q_k^{\text{sub}})}{M_{\text{ave}} \;-\; (s + Q_k^{\text{sub}})}, \tag{171}$$

provided $M_{\text{ave}} > s + Q_k^{\text{sub}}$ (average subset is better than equal weights in population, which holds for k close to m).

Proof. Immediate from $L_k^{\text{sub}} < L^{\text{ave}}$, cross-multiplying, and rearranging. □

11.4 Infinite-bagged RSM versus full estimated optimal weights

Proposition 4 (Exact infinite-bagged dominance)**.** *Infinite-bagged RSM beats full estimated optimal weights at subset size k iff*

$$\boxed{R_k^{\text{bag},\infty}\,(T-m) \;<\; A\,(m-k),} \tag{172}$$

where $R_k^{\text{bag},\infty} = \sum_{i=1}^{m}(\bar{v}_{i,k} - w_i^{\text{opt}})^2/\tau_i$ is the exact excess bagged risk from identity (166).

Proof. From $L_k^{\text{bag},\infty} < L^{\text{opt}}$, cancel $T-1$ and cross-multiply: $(s + Q_k^{\text{bag},\infty})(T-m) < (s + Q^{\text{opt}})(T-k)$. Substitute $Q_k^{\text{bag},\infty} = Q^{\text{opt}} + R_k^{\text{bag},\infty}$: $A(T-m) + R_k^{\text{bag},\infty}(T-m) < A(T-k)$. Rearranging gives (172). □

Remark 2 (Geometric interpretation)**.** *Condition (172) can be read as: the weighted distance between the bagged weight vector $\bar{v}_k$ and the oracle optimum w^{opt}, measured in the τ^{-1}-weighted ℓ^2 metric, must be small enough relative to how much estimation inflation the full-pool estimator suffers. The right-hand side $A(m-k)/(T-m)$ is the "estimation budget": it is large when T is close to m (many parameters relative to data) or when $m-k$ is large (bagging uses far fewer forecasters than the full pool). As $k \to m$, both sides go to zero and the condition is satisfied trivially (bagged = full optimum in population).*

Remark 3 (Monotonicity and the exact threshold)**.** *Because $R_k^{\text{bag},\infty}$ is decreasing in k (larger subsets yield bagged weights closer to w^{opt}) and $A(m-k)$ is also decreasing in k, the comparison is between two decreasing functions. The exact dominance threshold κ^{exact} is the smallest k satisfying (172), i.e. the solution to*

$$R_\kappa^{\text{bag},\infty}(T-m) = A(m-\kappa). \tag{173}$$

This can be located numerically by computing $R_k^{\text{bag},\infty}$ via Monte Carlo and checking (172) on the integer grid $k = 1, \ldots, m-1$.

Corollary 2 (Small-heterogeneity recovery). *Substituting* $R_k^{\text{bag},\infty} \approx C_\delta(1/k - 1/m)^2 = C_\delta(m-k)^2/(k^2m^2)$ *into (172) and dividing by* $(m-k) > 0$*:*

$$\frac{C_\delta(m-k)(T-m)}{k^2m^2} < A.$$

Rearranging: $Am^2k^2 > C_\delta(T-m)(m-k)$*, which is eq. (131) of the main text.*

11.5 Finite-bagged RSM versus full estimated optimal weights

Proposition 5 (Exact finite-bagged dominance). *For* G *independent draws, finite-bagged RSM beats full estimated optimal weights at subset size* k *iff*

$$\boxed{\left(R_k^{\text{bag},\infty} + \frac{Q_k^{\text{sub}} - Q_k^{\text{bag},\infty}}{G}\right)(T-m) \; < \; A\,(m-k).} \tag{174}$$

Proof. By Proposition 1 of the main text, $Q_k^{\text{bag},G} = Q^{\text{opt}} + R_k^{\text{bag},\infty} + (Q_k^{\text{sub}} - Q_k^{\text{bag},\infty})/G$. Apply the same argument as Proposition 4. □

The extra term $(Q_k^{\text{sub}} - Q_k^{\text{bag},\infty})/G$ is the finite-bagging penalty: it is positive (since $Q_k^{\text{sub}} \geq Q_k^{\text{bag},\infty}$), decreases in G, and vanishes as $G \to \infty$, recovering Proposition 4.

In the common-loading model, the exact values $Q_k^{\text{sub}} = q + E_S[1/A_S]$ and $Q_k^{\text{bag},\infty} = q + \sum_i \bar{v}_{i,k}^2/\tau_i$ are both estimable by Monte Carlo, so condition (174) is directly computable without any distributional assumption on $\{\tau_i\}$.

Corollary 3 (Small-heterogeneity recovery). *Substituting* $Q_k^{\text{sub}} - Q^{\text{opt}} \approx B(1/k - 1/m)$ *and* $R_k^{\text{bag},\infty} \approx C_\delta(1/k-1/m)^2$*, and using* $Q_k^{\text{sub}} - Q_k^{\text{bag},\infty} \approx B(1/k - 1/m) - C_\delta(1/k-1/m)^2$*, condition (174) reduces (after dividing by* $m - k > 0$ *and rearranging) to*

$$Am^2k^2 > (T-m)\big[a_G mk + c_G(m-k)\big],$$

which is eq. (138) of the main text.

11.6 Infinite-bagged RSM versus equal weights

Proposition 6 (Exact infinite-bagged vs equal-weights dominance). *Infinite-bagged RSM beats equal weights at subset size* k *iff*

$$\boxed{(A + R_k^{\text{bag},\infty})(T-1) \; < \; M_{\text{ave}}\,(T-k).} \tag{175}$$

Equivalently,

$$R_k^{\text{bag},\infty}(T-1) < (M_{\text{ave}} - A)(T-k) - A(k-1). \tag{176}$$

Proof. From $L_k^{\text{bag},\infty} < L^{\text{ave}}$: $(s + Q_k^{\text{bag},\infty})\frac{T-1}{T-k} < s + Q^{\text{ave}}$. Multiply both sides by $T - k > 0$ and substitute $Q_k^{\text{bag},\infty} = Q^{\text{opt}} + R_k^{\text{bag},\infty}$: $(A + R_k^{\text{bag},\infty})(T-1) < M_{\text{ave}}(T-k)$. Equation (176) follows by expanding $A(T-1) = A(T-k) + A(k-1)$ and subtracting. □

Remark 4 (Sufficient conditions for dominance over equal weights). *Condition (176) is satisfied if two sub-conditions hold simultaneously: (i) $R_k^{\mathrm{bag},\infty}$ is small (bagged weights close to oracle optimum), and (ii) $M_{\mathrm{ave}} - A = Q^{\mathrm{ave}} - Q^{\mathrm{opt}}$ is large (equal weights substantially worse than the oracle). For large T, the leading constraint simplifies to $R_k^{\mathrm{bag},\infty} < Q^{\mathrm{ave}} - Q^{\mathrm{opt}}$, i.e. $Q_k^{\mathrm{bag},\infty} < Q^{\mathrm{ave}}$—the population bagged risk must be below the equal-weight population risk.*

11.7 Summary: exact versus approximate conditions

Table 1 collects the exact conditions alongside their small-heterogeneity counterparts from Section 10 of the main text.

Table 1: Exact dominance conditions under common loading (no additional assumptions) versus their small-heterogeneity approximations. All conditions require $T > m$ when comparing against full estimated optimal weights. Here $\mathcal{E}_k = Q_k^{\mathrm{sub}} - Q^{\mathrm{opt}}$, $z_k = 1/k - 1/m$.

Comparison	Exact condition	Small-het approximation
Avg-sub vs. L^{opt}	$\mathcal{E}_k(T-m) < A(m-k)$	$k > B(T-m)/(mA)$
Avg-sub vs. L^{ave}	$(s + Q_k^{\mathrm{sub}})(T-1) < M_{\mathrm{ave}}(T-k)$	quadratic in k (eq. 124)
∞-bag vs. L^{opt}	$R_k^{\mathrm{bag},\infty}(T-m) < A(m-k)$	$Am^2k^2 > C_\delta(T-m)(m-k)$
Finite-bag vs. L^{opt}	$(R_k^{\mathrm{bag},\infty} + \frac{\Delta_k}{G})(T-m) < A(m-k)$	$Am^2k^2 > (T-m)[a_G mk + c_G(m-k)]$
∞-bag vs. L^{ave}	$(A + R_k^{\mathrm{bag},\infty})(T-1) < M_{\mathrm{ave}}(T-k)$	cubic in k (eq. 144)

$$\Delta_k = Q_k^{\mathrm{sub}} - Q_k^{\mathrm{bag},\infty} \geq 0.$$

11.8 How to use the exact conditions in simulations

The exact conditions depend only on quantities that are directly estimable by the Monte Carlo design described in Section 11 of the main text:

1. Estimate $Q_k^{\mathrm{sub}} = q + E_S[1/A_S]$ by averaging $1/A_S$ over Monte Carlo subset draws. This is already reported as `Q_subset_mc` in the exogenous simulation code.
2. Estimate $Q_k^{\mathrm{bag},\infty} = q + \sum_i \bar{v}_{i,k}^2/\tau_i$ by accumulating the averaged weights $\bar{v}_{i,k}$ over subset draws. This is already reported as `Q_bag_inf_mc`.
3. Compute $R_k^{\mathrm{bag},\infty} = Q_k^{\mathrm{bag},\infty} - Q^{\mathrm{opt}}$ (reported as `R_bag_inf_mc`) and check conditions (172) and (174) over the integer grid $k = 1, \ldots, m-1$.
4. The exact dominance threshold κ^{exact} is the smallest k for which the condition holds. Compare it against the small-heterogeneity threshold from eq. (133) to assess the quality of the approximation.

The main advantage of the exact approach is that it does not require small heterogeneity: it applies to any common-loading design, including cases with large $H = \bar{\sigma}^2\bar{\tau}$ or strongly skewed $\{\tau_i\}$, which are relevant for empirical forecast panels.

Connection to the endogenous case. In the endogenous code, the forecast error covariance Σ_e is no longer common-loading (it is Toeplitz or factor-structured). The exact identity (166) becomes $R_k^{\text{bag},\infty} = (\bar{v}_k - w^{\text{opt}})'\Sigma_e(\bar{v}_k - w^{\text{opt}})$, which remains valid for any positive-definite Σ_e and is estimated via `q_population_mc_general` in the simulation. However, the closed-form approximations for $\bar{v}_k$ derived in Section 7 no longer apply because the subset weights are not of the form $w_{i,S} = \tau_i/A_S$. The dominance conditions (172)–(175) remain formally valid with this generalized $R_k^{\text{bag},\infty}$, but computing the threshold requires numerical Monte Carlo rather than the analytical formulas of Section 10.